



Chapter 10

iSeq: Web-Based RNA-seq Data Analysis and Visualization

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Abstract

Transcriptome sequencing (RNA-seq) is becoming a standard experimental methodology for genome-wide characterization and quantification of transcripts at single base-pair resolution. However, downstream analysis of massive amount of sequencing data can be prohibitively technical for wet-lab researchers. A functionally integrated and user-friendly platform is required to meet this demand. Here, we present iSeq, an R-based Web server, for RNA-seq data analysis and visualization. iSeq is a streamlined Web-based R application under the Shiny framework, featuring a simple user interface and multiple data analysis modules. Users without programming and statistical skills can analyze their RNA-seq data and construct publication-level graphs through a standardized yet customizable analytical pipeline. iSeq is accessible via Web browsers on any operating system at <http://iseq.cbi.pku.edu.cn>.

Key words RNA-seq, R-Shiny, Gene expression analysis, Gene ontology enrichment, Data visualization

1 Introduction

Next-generation sequencing (NGS) technologies have been playing an essential role in the studies on genomics, transcriptomics, and epigenomics in the recent years. Their ability of sequencing multiple nucleic acid molecules in parallel makes it possible to generate large datasets and thus offer new insights to many biological questions [1]. Transcriptome sequencing (RNA-seq) utilizes NGS technologies for determination and quantification of RNA molecules in a biological sample. It provides higher coverage and improved sensitivity for genome-wide expression profiling compared to previous methods represented by microarray assays and has gained immense popularity due to ever-increasing through-

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Tao Huang (ed.), *Computational Systems Biology: Methods and Protocols*, Methods in Molecular Biology, vol. 1754, https://doi.org/10.1007/978-1-4939-7717-8_10, © Springer Science+Business Media, LLC, part of Springer Nature 2018

put and decreasing cost. The rise of RNA-seq methodologies has greatly deepened our understandings of embryonic development [2], carcinogenesis [3], cell differentiation [4], and many other research areas.

1.1 An Overview of RNA-seq Workflow

A complete RNA-seq procedure consists of both experimental stage and analysis stage. Although several sequencing protocols exist for RNA-seq, general steps and outputs in the experimental stage are similar. Briefly, RNA molecules with poly-A tails are first isolated by oligo-dT priming [5]. Alternatively, non-rRNAs are enriched by rRNA depletion [6]. The resulting RNAs are fragmented and then reverse-transcribed into short (200–1000 bp) cDNA fragments, which are then attached with sequencing adaptors and sequenced from one end or both ends. Several NGS technologies, including Illumina [7] and SOLiD [8], can be used for RNA-seq to generate millions or billions of short reads representing DNA segments.

The analysis stage of RNA-seq begins with mapping reads to the reference genome. Because eukaryotic genomes contain introns, RNA-seq reads often have gaps with varying lengths up to hundreds of thousands of base pairs, which make DNA sequence mapping tools generally unsuitable for direct use in RNA-seq. Widely used RNA-seq mapping tools include Tophat [9], SOAP [10], and GSNAP [11]. There are also programs that map reads onto a reference transcriptome, rather than reference genome, to circumvent the gap problem and to reduce computation time, such as Sailfish [12] and Kallisto [13]. Following read mapping is the quantification of each RNA species that are either provided by the reference transcriptome or de novo assembled from reads. Most mapping software packages also perform the quantification step. Generally, the final output of mapping and quantification steps can be described as a matrix with each column being a sample and each row being a gene or a splicing isoform of a transcript. This matrix, often called the expression profile, is the starting point of downstream analysis of RNA-seq datasets.

The expression profile contains rich transcriptomic information regarding the tested samples. How to draw biological meanings from it, however, is highly contingent on the specific research background. A one-size-fits-all analytical workflow does not exist. For example, different normalization methods have been proposed to alleviate technical variations and batch effects among samples, each with its strengths and drawbacks [14]. Theoretically predicting which method will give best results is a challenging or sometimes impossible task [14]. RNA-seq data are often analyzed by clustering methods to discover co-expressed gene groups or sample subclasses that share expression patterns. Commonly used clustering algorithms, including hierarchical clustering, principal

component analysis (PCA) [15], and t-distributed stochastic neighbor embedding (t-SNE) [16], are each applicable for different data types, sample numbers, and research questions.

1.2 Web-Based RNA-seq Data Analysis

There is a notable contrast of skill requirement between the experimental stage and analysis stage of RNA-seq assays. The former needs experimental operation skills in cell culturing, RNA isolation, and library preparation and can be readily done in most traditional wet laboratories. However, the required computational and programming skills in the analysis stage are often less familiar to most biologists. There are commercial companies providing NGS data analysis services, but their high costs, inefficient information sharing, and communication delays are often not satisfying.

Web-based bioinformatics tools are now emerging to alleviate the situation. Some of them, exemplified by the Galaxy [17] and Seven Bridges (www.sbgenomics.com), generally aim at next-generation sequencing (NGS) data processing and are not specialized for RNA-seq analysis. Galaxy utilizes a graphical workflow editor to allow users to conduct their genomic data analysis workflow with interactivity and extensibility. However, it requires users to be familiar with each tool to achieve expected results and has many limitations including storage space, data transfer speed, and maximum job submissions. The Seven Bridges contains a suite of more than 200 pipelines and applications to help interpret bioinformatics data, emphasizing the security of data and analysis results. However, it is a commercial Web site that charges storage and computation costs and also requests high level of bioinformatics skills for proper usage.

Web-based tools specifically designed for RNA-seq include START [18], RAP [19], and CANEapp [20]. These tools provide more targeted solutions to extract information from RNA-seq datasets. RAP is a free cloud computing application with a fully automated and standardized pipeline dedicated mainly to read mapping, quantification, alternative splicing, and RNA editing detection. CANEapp shares many features with RAP except that it mainly focuses on detection of differential gene expression and novel noncoding RNA. START is an open-source application which can be run both locally and on the server side. It is user-friendly to wet-lab researchers and provides data visualization. However, this tool only provides basic visual interpretations of input datasets, such as heatmap, box plot, and volcano plot. Many important tasks within RNA-seq analysis, including normalization, differential expression detection, and functional enrichment, are lacking in this tool.

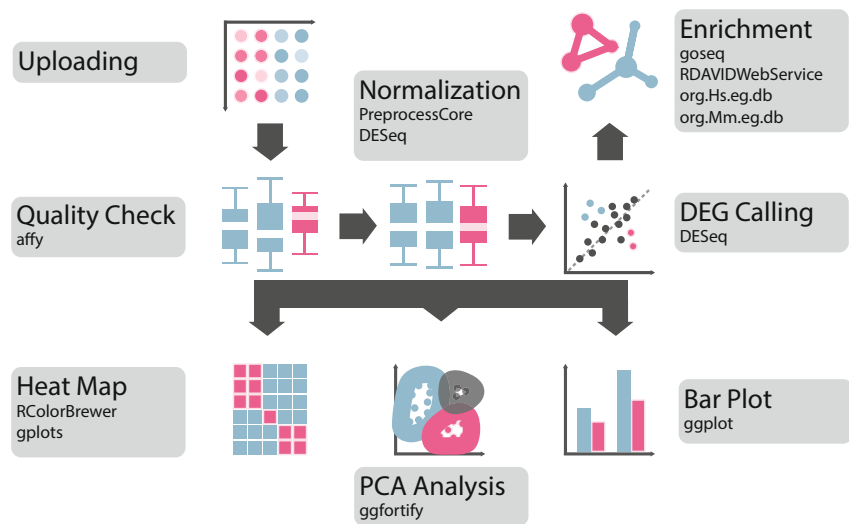


Fig. 1 An overview of iSeq analysis workflow

Table 1
Comparison between iSeq and other Web-based RNA-seq analysis tools

	iSeq	Galaxy	Seven Bridges	START	RAP	CANEapp
Local installation	Yes	Yes	No	Yes	No	No
Skill requirements	Low	High	High	Low	Middle	Middle
Mapping	No	Yes	Yes	No	Yes	Yes
RNA editing	No	Yes	Yes	No	Yes	No
Quality check ^a	Yes	No	No	No	No	No
Normalization	Yes	Yes	Yes	No	Yes	Yes
DEG calling	Yes	Yes	Yes	No	Yes	Yes
Visualization	Rich	Basic	Basic	Rich	Basic	No

^aQuality check here refers to checking the comparability among samples which is susceptible to batch effects. Not to be confused with checking read quality as is often done by FastQC and similar tools

1.3 iSeq: A Web-Based Server for RNA-seq Data Analysis and Visualization

In this article, we introduce iSeq, a Web-based analysis suite specially designed for RNA-seq data analysis, which integrates robust analysis methods, data visualization schemes, and functional analysis tools to analyze and visualize RNA-seq data (Fig. 1). Users can explore their data through an intuitive graphical interface and focus on deciphering expressional patterns and functional relevance central to their experimental questions, without having to master complex computational and programming skills.

A comparison between iSeq and other Web-based RNA-seq analysis tools is summarized in Table 1. The greatest distinction of iSeq lies in that it focuses on a complete analytical workflow

downstream of gene-expression profile. RNA-seq workflow involves a wide range of bioinformatics tools and requires a high level of computational skills to accomplish manually. Moreover, the interaction among different analysis steps brings additional workload and complexity. For example, the choice of normalization method will have impact on nearly all downstream results and may need to be re-performed after downstream quality-checking and clustering analysis. However, these tasks consume a very small amount of time, which makes them particularly suitable to be integrated in an interactive graphical user interface. Unlike most other tools (except START), iSeq is a lightweight application that makes it possible to complete all tasks within an hour (*see* **Note 1**).

2 Materials

In this section, we describe the software, packages, and methods used in building the iSeq Web server. The implementation of iSeq was based on Shiny, an open-source R package for turning R analyses into interactive Web applications that are easy to use. There are five modules in iSeq—data uploading, normalization, differentially expressed genes (DEGs) calling, functional enrichment, and plots. These modules realize a complete analysis pipeline starting from the gene-expression profile. Each module integrates a set of R packages that are key to its functioning, as listed below.

There are two available methods in the normalization module to normalize the input dataset. The size factor method is implemented in DESeq, an R package for differential RNA-seq analysis [21]. This package is also used in the DEG calling module to detect differentially expressed genes. The quantile method was initially developed to normalize microarray datasets but has also been shown effective to normalize RNA-seq data. This method is incorporated into the R package “preprocessCore” (<https://github.com/bmbolstad/preprocessCore>).

The functional enrichment module integrates multiple gene functional enrichment methods to facilitate a comprehensive functional analysis, revealing the biological meaning behind a select group of genes. DAVID [22, 23] is a widely used online Web server that provides functional enrichment analysis of a list of genes using gene ontology (GO) and pathway information. GOSec [24], an R package that performs GO analysis, is also available in this module.

iSeq leverages powerful graphing packages in R to construct high-quality figures for visualization and publication. Most figure outputs in iSeq are produced by ggplot2 [25], an R package for providing beautiful plots while taking care of plotting details to meet individualized requirements. Several statistical plots including principal component analysis (PCA) are supported by the “ggfortify” R package [26]. Specialized color schemes in heatmaps and

other plots are realized by the “RColorBrewer” R package (<https://cran.r-project.org/web/packages/RColorBrewer/RColorBrewer.pdf>).

3 Methods

In this section, we will show how to analyze RNA-seq expression data using iSeq. We reanalyzed a public RNA-seq data from Gene Expression Omnibus (GEO accession: GSE39866), which was published on *Nature Neuroscience* [27]. This dataset reported the mRNA expression differences between embryonic and adult mouse cerebral cortex and identified several genes (*Mobp*, *Igf2bp* et al.) involved in important pathways (ion transport, cell cycle) with altered expression during the cerebral cortex developmental processes. Using iSeq, we could easily reproduce the same results as in the paper without using command line analysis or programming.

3.1 Prepare Data

In the *Nature Neuroscience* paper, the authors generated seven RNA-seq samples for two biological conditions, consisting of four replicate samples for embryonic and three replicate samples for adult mouse cerebral cortex.

1. Expression File of Genes

Download the expression data file from <http://202.205.131.33:3838/expression.csv>. In this file, each row represents a certain gene, and each column represents a certain sample (Fig. 2). The value of each entity represents the expression level as measured by raw sequence read counts; the FPKM and TPM are also acceptable. The first column and row list the names of genes and samples, respectively.

2. Condition File of Samples

Download the sample description file from <http://202.205.131.33:3838/condition.csv>. This file allows classifying samples into biological conditions. It has two rows, with the first row listing sample names and the second row listing condition names (Fig. 2). Make sure to use the same sample names in the expression file and condition file.

3.2 Upload Data

There are two ways to access iSeq: either visit the online version (<http://iseq.cbi.pku.edu.cn>) or install a local version of iSeq on your own computer (<https://github.com/ChengLiLab/iSeq>). Figure 2 shows the interface to upload the expression table (expression.csv) and the sample condition table (condition.csv).

3.3 Normalization

Some systematic variations have been reported, such as the library size, gene length, and GC content [28]. Normalization is an import step for RNA-seq to remove these systematic biases. After

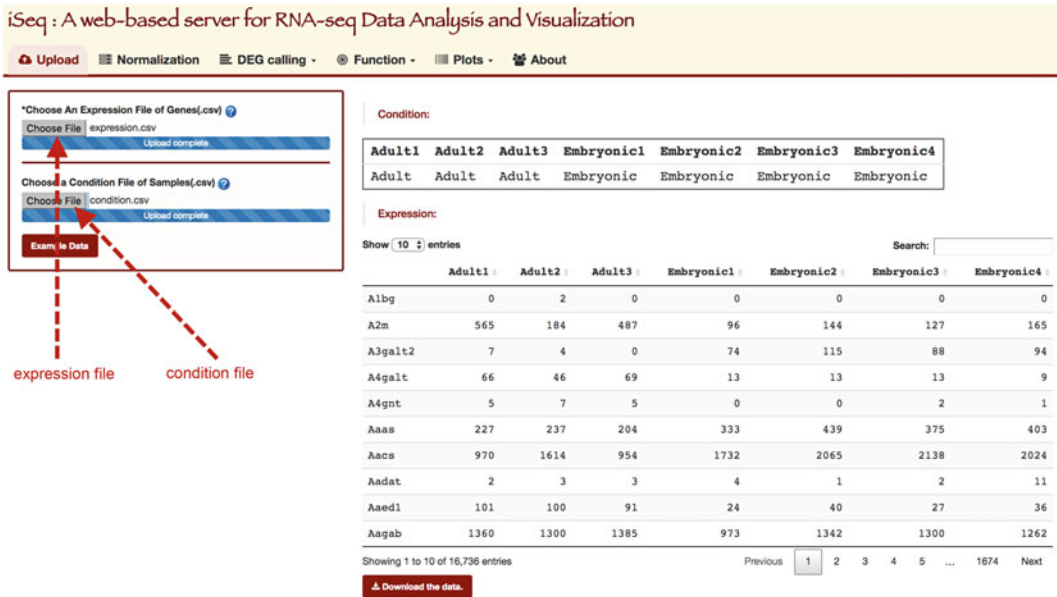


Fig. 2 The iSeq data uploading page. The arrows show the uploading buttons for expression file and sample condition file. The main page on the right shows the previews of the uploading files

normalizing the data, the effectiveness can be evaluated by examining the comparability of the gene expression distribution across samples (box plot) and the similarity among samples within each condition (hierarchical clustering).

iSeq provides two widely used normalization methods, quantile normalization and size factor. In this example, we choose size factor normalization method (Fig. 3). The plots on this page will automatically update, when user chose a new normalization method.

1. Box Plot

In this plot, each box represents the distribution of gene expression levels of a sample (Fig. 3). Well-normalized expression profiles have expression patterns with similar distribution among samples.

2. Hierarchical Clustering

Hierarchical clustering outputs a tree structure to visualize similarity relationships among samples (Fig. 4). Here we use the genes whose average expression values are higher than one to calculate the distance among samples. The height of a branching point stands for the similarity among samples in the subtree below it, with more similar samples having lower branching points connecting them. As expected, the sample clustering corresponds well with the partitioning by biological conditions. In this example, the four embryonic samples and three adult samples are clustered together, respectively (Fig. 4).

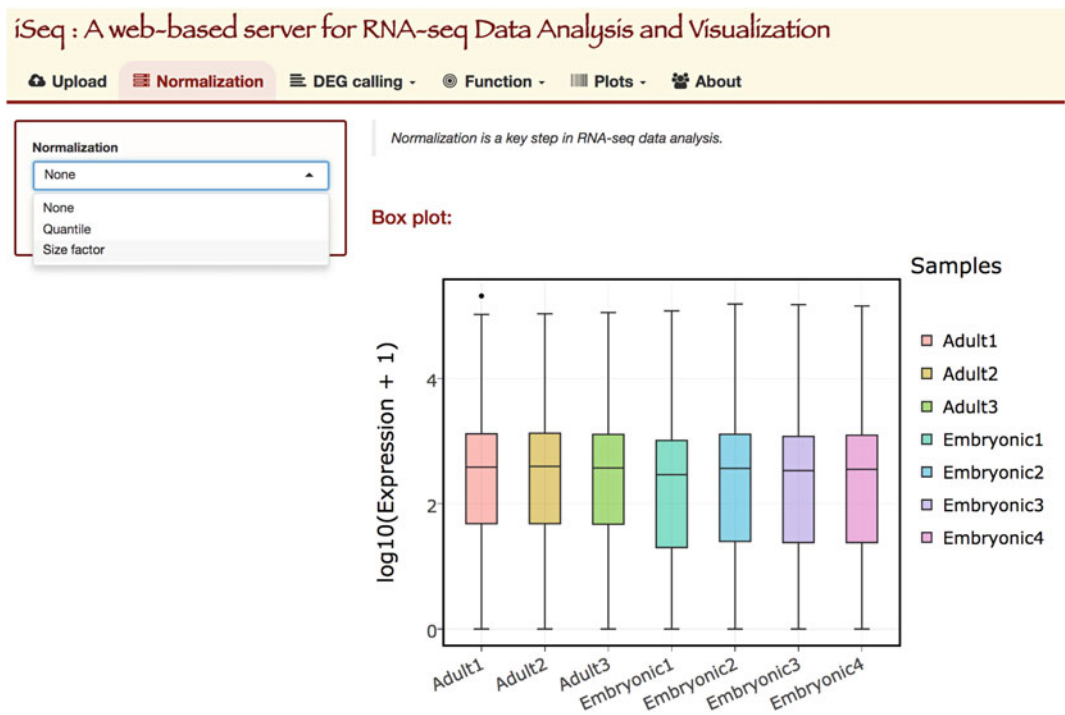


Fig. 3 Size factor normalization for the RNA-seq data. The box plot on the right shows the distribution of gene expression values in each sample. The box plot is a standardized way of displaying the distribution of a set of data points. The central rectangle spans from the first quartile to the third quartile (the 25th to 75th percentiles), and this distance is termed interquartile range (IQR). A segment inside the rectangle shows the median, and whiskers stretching outside the box show the 1.5*IQR from the lower and upper quartiles. Data points beyond the end of the whiskers are called “outlier points” and are plotted individually

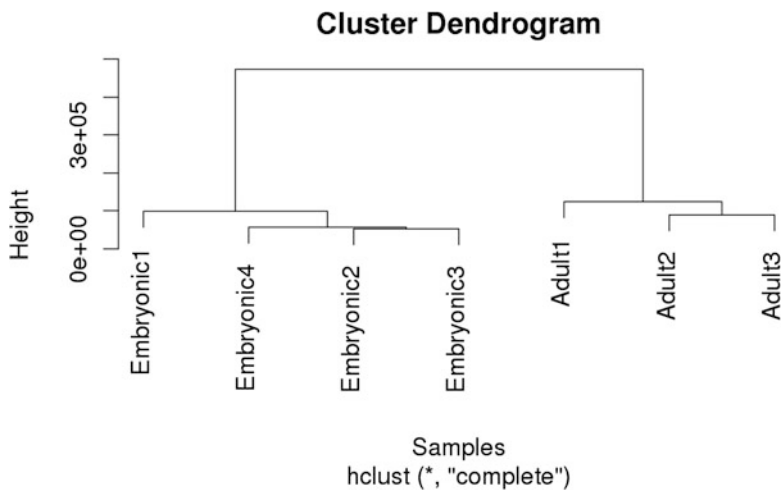


Fig. 4 Hierarchical clustering for all samples. The replicate embryonic and adult samples cluster together, respectively

3.4 Detect Differentially Expressed Genes (DEGs)

1. We use the R package, DESeq, to detect differentially expressed genes between two biological conditions. DESeq has great consistency in terms of the output gene list when the number of replicates is as small as 2–5. When the number of replicates increases to beyond 10, it also produces low false-positive rates [29].

There are several parameters to run the DEG module:

- (a) padj cutoff: The individual p -value for each gene after being adjusted for multiple testing with the Benjamini-Hochberg procedure. Genes with smaller padj are regarded to be differential expressed with higher statistical significance. Setting a smaller cutoff value will result in a more stringent test and fewer DEG genes. The default padj is 0.05.
 - (b) Fold-change cutoff: The fold change is defined as the ratio of mean gene expression values under two conditions. The greater the relative difference, the further fold change departs from 1. Setting a larger cutoff value will result in a more stringent test and fewer DEG genes and vice versa. Here we chose “less than 0.25 or greater than 4,” which is used in the original paper of this dataset.
 - (c) Base mean cutoff: The mean expression value of a gene among all samples under both conditions. This filter is intended to remove genes with very low expression, which often leads to unreliable large fold-change values. Here, we set the value to 10, which means, if a gene covers less than 10 reads on average, it will be not be called as a DEG.
2. Click the “Run to detect” button to start running. DEG calling is a time-consuming step in the RNA-seq data analysis pipeline. This step takes about 3 min. When it is finished, we will get a differentially expressed gene list (Fig. 5) and a volcano plot (Fig. 6), which is widely used in RNA-seq analysis to identify DEGs (upper-left and upper-right areas in the plot). This DEG list can be downloaded as a CSV file to be viewed or analyzed by other software.

3.5 Reveal the Biological Meaning Behind DEGs

1. Click the Function menu and select the Online servers (Fig. 7).
2. Copy the upregulated or downregulated gene list to clipboard.
3. Click the “David,” and go to the official site of The Database for Annotation, Visualization and Integrated Discovery (DAVID, <http://david.ncifcrf.gov>).
4. Gene ontology and pathway enrichment analysis using DAVID.
 - (a) Paste the upregulated or downregulated gene list.
 - (b) Select gene identifier; the gene in the example list is “OFFICAL_GENE_SYMBOL”. Make sure you select

Show 10 entries

Search:

	baseMean	baseMeanA	baseMeanB	foldChange	log2FoldChange	pval	padj
Cd24a	12143.4285714286	222.666666666667	21084	94.688622754491	6.56511918497664	0	0
Clu	8015.71428571429	17869.3333333333	625.5	0.0350041038650948	-4.83633211689594	0	0
Cntnap1	2996.42857142857	6888.66666666667	77.25	0.0112140714216588	-6.47854602640193	0	0
Cx3c1l	7407.42857142857	16568.3333333333	536.75	0.0323961372095363	-4.94803438780291	0	0
Ddn	12609	28969	339	0.0117021643826159	-6.41708080054019	0	0
Dpysl3	34964.1428571429	1347.33333333333	60176.75	44.6635947550717	5.48102746621359	0	0
Dpysl5	17893.1428571429	1020.33333333333	30547.75	29.9389905259719	4.90395367273905	0	0
Draxin	2923.85714285714	37.6666666666667	5088.5	135.092920353982	7.07780826090149	0	0
Enpp2	5468.42857142857	12486.3333333333	205	0.0164179502923196	-5.92858216544151	0	0
Faim2	6308.71428571429	13873.6666666667	635	0.0457701640998534	-4.4494487253297	0	0

Showing 1 to 10 of 2,556 entries

Previous

1

2

3

4

5

...

256

Next

Fig. 5 The differentially expressed genes list between embryonic and adult samples. Each row represents a DEG, and columns contain the mean expression (base mean), the expression fold change, and the adjusted *p*-value (padj)

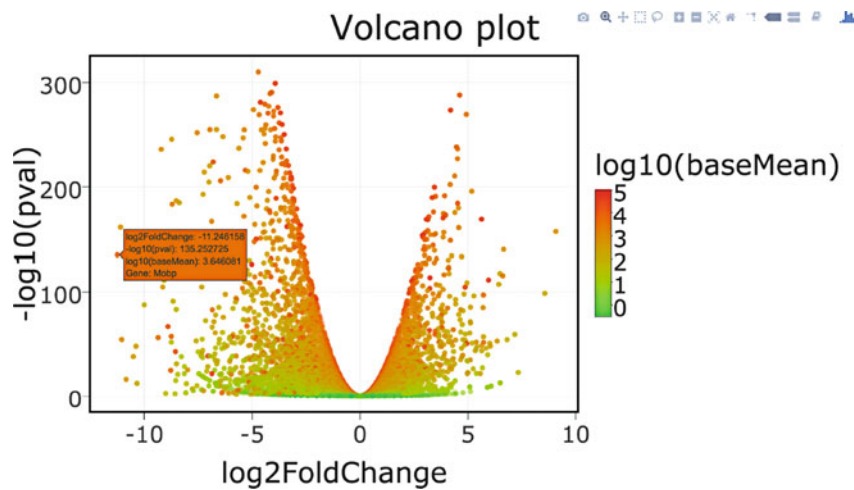


Fig. 6 The volcano plot shows the log *p*-value against the logarithm of fold change. Each dot represents a gene, with the color indicating the logarithm of mean expression of the gene. Pointing the mouse to a dot displays the supplementary information for the gene. The example is the supplementary information for *Mobp*

the right gene identifier. Take the gene “myelin-associated oligodendrocyte basic protein” as an example. The official gene symbol is *Mobp*, the HGNC gene id is “7189,” the Entrez gene id is “4336,” and the Ensemble gene id is “ENSG00000168314.”

- (c) Select gene list, which means that we upload a gene list.
- (d) Submit the list.
- (e) Select species.

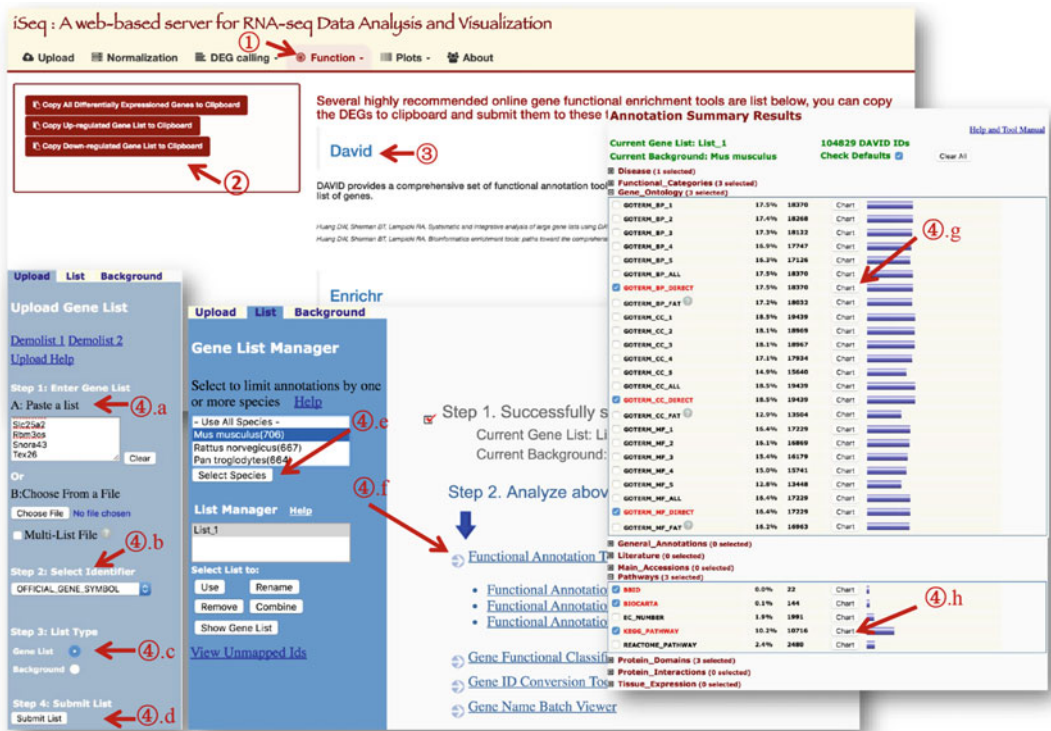


Fig. 7 The pipeline for gene enrichment analysis using iSeq and DAVID. The numbers and letters show the order of the enrichment analysis steps

- (f) Analyze the uploaded gene list with “function annotation tools.”
- (g) Open the “Gene_Ontology” expanding menu and select the chart of “GOTERM_BP_DIRECT.” This chart shows the enrichments of gene ontology terms in biological process. The results are similar to the original paper of the example data, with upregulated genes enriched in “cell cycle” and “cell division” and downregulated genes enriched in “ion transport” and “potassium ion transport.”
- (h) Open the “Pathways” expanding menu and select the chart of “KEGG_PATHWAY.” This chart shows the enrichment KEGG pathways in the input gene list.

3.6 Frequently Used Plots in RNA-seq Analysis

We integrate some frequently used plots in the “plots” module, including the bar plot, heatmap, PCA and t-SNE.

1. Gene expression bar plot. The bar plot shows the expression level of a given gene in all samples, grouped by conditions. As showed in Fig. 8a, gene *Mobp* has much higher expression in adult samples than in embryonic samples.

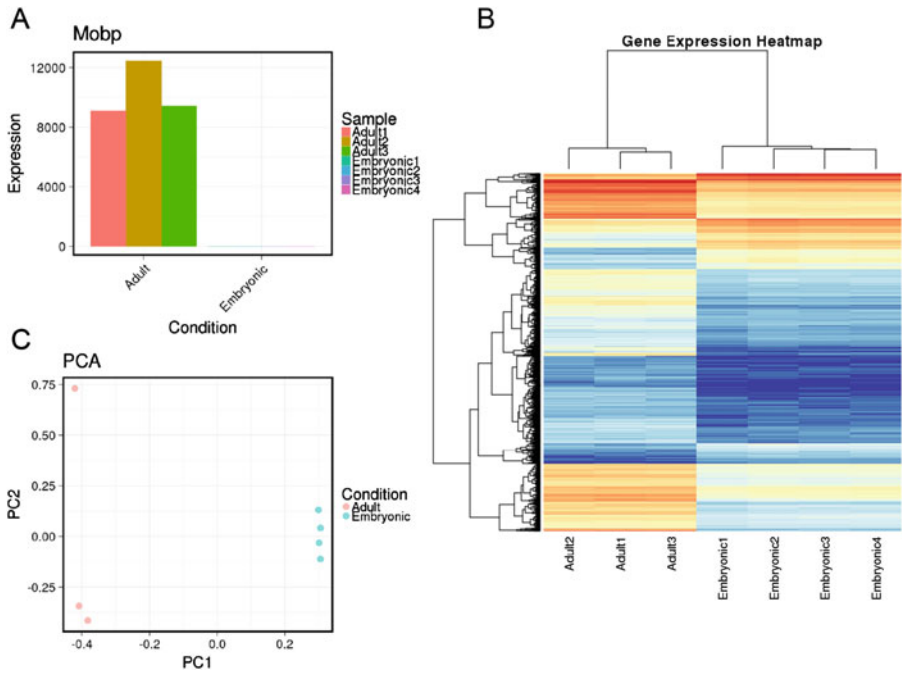


Fig. 8 (a) The expression levels of Mobp gene in all samples. The expression levels in adult are much higher than that in embryonic samples. (b) Heatmap for DEGs between embryonic and adult samples. (c) Principal component analysis shows that the adult and embryonic samples are separated into two clusters, indicating they are in two different conditions

2. Heatmap. In the gene expression heatmap (Fig. 8b), each row represents a gene, and each column represents a sample. Rows and columns are hierarchically clustered. Gene sets with specific expression patterns can be identified from the heatmap. The default genes used in the heatmap are the DEGs called by the “DEG calling” module; user can also upload other genes through the left menu to plot the heatmap.
3. Principal component analysis (PCA). PCA projects high-dimensional data points onto a low-dimensional space for visualization. The orthogonal axes of the space are names PC1 (principal component 1), PC2, and so on. They are chosen in such a way that the projected data points have the largest variance in the direction of PC1 and the second largest in PC2. The overall relationship or clustering of data points in the original high dimension can be visualized and identified in the low dimension more easily. In this example, the embryonic and adult samples are separated into two clusters, which are similar to the result from hierarchical clustering (Fig. 8c).

4 Notes

1. iSeq provides great flexibility in usage in that it is not required to run each module sequentially. For example, the DEG calling step could be skipped if a gene set other than DEGs is used in the enrichment module. Although iSeq is designed for NGS data analysis, datasets from Sanger sequencing or microarray assays, if converted to the required input format, could also be analyzed on this platform.

As a lightweight RNA-seq data analysis application, iSeq imposes short waiting period between user operation and result display. Users generally complete all analysis tasks within an hour, without needing the requirement of registration. Currently, iSeq does not perform read mapping because it consumes massive computational resources and lasts much longer than other tasks. Researchers may resort to mapping software (e.g., Tophat) or online tools (e.g., Galaxy) for read alignment and then seamlessly shift to the iSeq workflow.

iSeq is an ongoing project on which further enhancement and extension will be our future effort. In addition to providing a broader range of choices in each module, we will also introduce a mapping functionality and a cloud storage system to realize a one-stop analytical pipeline. We also notice that biologists with output data from other sequencing technologies such as bisulfite sequencing (BS-seq) and whole genome sequencing (WGS) are facing similar difficulties as with RNA-seq and will possibly extend iSeq into these areas.

Acknowledgments

We thank Yifang Liu for advice on Web server construction and the PKU Bioinformatics Core Discussion Group (Yangchen Zheng, Yong Peng) for testing and suggestions. This work was supported by funding from Peking-Tsinghua Center for Life Sciences and School of Life Sciences of Peking University, Natural Science Foundation of China (Key Research Grant 71532001), and Chinese National Key Projects of Research and Development (2016YFA0100103).

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